CSC 370 PaperTrade, Functional Dependencies and Candidate Keys

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Overview

This document gives the set of functional dependencies (FDs) for each relation in the `stock_trading` database (`user`, `stock`, `transaction` and `holds`). For each relation it computes attribute closures (X^+), deriving every step from Armstrong's axioms, and uses them to find all candidate keys and superkeys. For each candidate key it shows that no proper subset is a superkey, so the key is minimal.

1 Armstrong's Axioms and Derived Rules

Let X, Y and Z be sets of attributes, and write XY for $X \cup Y$.

- **(A1) Reflexivity:** if $Y \subseteq X$, then $X \to Y$.
- **(A2) Augmentation:** if $X \to Y$, then $XZ \to YZ$.
- **(A3) Transitivity:** if $X \to Y$ and $Y \to Z$, then $X \to Z$.

Two more rules follow from these and are used in the derivations below.

- **(U) Union:** if $X \to Y$ and $X \to Z$, then $X \to YZ$.

  Proof. Augmenting $X \to Y$ with X gives $X \to XY$ (A2, since $XX = X$). Augmenting $X \to Z$ with Y gives $XY \to YZ$ (A2). By transitivity, $X \to YZ$ (A3).

- **(D) Decomposition:** if $X \to YZ$, then $X \to Y$.

  Proof. $YZ \to Y$ by reflexivity (A1), so $X \to Y$ by transitivity (A3).

Closures, Superkeys and Candidate Keys

For a relation R with attribute set C and FD set F, the **closure** X^+ is the set of all attributes A such that $X \to A$ follows from F by Armstrong's axioms. Then:

- X is a **superkey** of R if and only if $X^+ = C$.
- X is a **candidate key** of R if it is a superkey and no proper subset of X is a superkey.

No-fire lemma. If X contains the left-hand side of no FD in F, then $X^+ = X$. The axioms only ever derive $X \to A$ for $A \in X$ (reflexivity), or for A on the right-hand side of an FD whose left-hand side is already in the closure. In particular, $\emptyset^+ = \emptyset$, since none of our FDs has an empty left-hand side.

2 `user`

$$C_U = \{\text{user_id}, \text{name}, \text{email}\}$$

FD Set F_U

- FD_1: user_id $\to$ name, email. Each user ID belongs to one account.
- FD_2: email $\to$ user_id. An email address can register only one account.

Dependencies that do not hold: name $\not\to$ user_id, since two users can share a name (two people named Alice Martin can both register).

Closure of {user_id}

1.	user_id $\to$ user_id	Reflexivity (A1)
2.	user_id $\to$ name, email	Given (FD_1)
3.	user_id $\to$ user_id, name, email	Union (U) of 1 and 2

$$\{\text{user_id}\}^+ = \{\text{user_id}, \text{name}, \text{email}\} = C_U$$

So {user_id} is a superkey.

Closure of {email}

1.	email $\to$ email	Reflexivity (A1)
2.	email $\to$ user_id	Given (FD_2)
3.	user_id $\to$ name, email	Given (FD_1)
4.	email $\to$ name, email	Transitivity (A3) of 2 and 3
5.	email $\to$ name	Decomposition (D) of 4
6.	email $\to$ user_id, name, email	Union (U) of 1, 2 and 5

$$\{\text{email}\}^+ = \{\text{user_id}, \text{name}, \text{email}\} = C_U$$

So {email} is a superkey.

Minimality

The only proper subset of {user_id} or {email} is $\emptyset$, and $\emptyset^+ = \emptyset \neq C_U$ by the no-fire lemma. So neither key can be reduced, and both are candidate keys.

No Other Candidate Keys

The FD left-hand sides are {user_id} and {email}. A set containing neither is a subset of {name}, and by the no-fire lemma $\{\text{name}\}^+ = \{\text{name}\} \neq C_U$. Every superkey therefore contains user_id or email, so it contains one of the two candidate keys.

Candidate keys: {user_id} and {email}, with {user_id} chosen as the primary key.

Superkeys (6): every superset of a candidate key: {user_id}, {email}, {user_id, name}, {email, name}, {user_id, email}, {user_id, name, email}.

3 `stock`

$$C_S = \{\text{ticker}, \text{company_name}, \text{opening_price}, \text{closing_price}, \text{exchange}\}$$

FD Set F_S

- FD_3: ticker → company_name, opening_price, closing_price, exchange. A ticker identifies one listed security, with one listing exchange and one session's prices.

Dependencies that do not hold:

- company_name ↛ ticker: one company can list several share classes under different tickers (for example, Alphabet trades as both GOOGL and GOOG).
- exchange ↛ ticker: each exchange lists many stocks.
- Prices determine nothing: two stocks can open or close at the same price.

Closure of {ticker}

1.	ticker → ticker	Reflexivity (A1)
2.	ticker → company_name, opening_price, closing_price, exchange	Given (FD_3)
3.	ticker → ticker, company_name, opening_price, closing_price, exchange	Union (U) of 1 and 2

$$\{\text{ticker}\}^+ = C_S$$

So {ticker} is a superkey.

Minimality

The only proper subset of {ticker} is $\emptyset$, and $\emptyset^+ = \emptyset \neq C_S$. So {ticker} is a candidate key.

No Other Candidate Keys

{ticker} is the only FD left-hand side. Any set X without ticker fires no FD, so $X^+ = X \neq C_S$ by the no-fire lemma. Every superkey therefore contains ticker.

Candidate key: {ticker} (the primary key).

Superkeys ($2^4 = 16$): $\{\text{ticker}\} \cup Y$ for every $Y \subseteq \{\text{company_name}, \text{opening_price}, \text{closing_price}, \text{exchange}\}$.

4 `transaction`

$$C_T = \{\text{transaction_id}, \text{user_id}, \text{ticker}, \text{transaction_type}, \text{quantity}, \text{price}, \text{time}\}$$

FD Set F_T

- FD_4: transaction_id → user_id, ticker, transaction_type, quantity, price, time. Each transaction ID records one trade.

Dependencies that do not hold: user_id, ticker, time ↛ transaction_id. `time` has one-second resolution, and a user can place two orders for the same stock within the same second (for example, a buy and an immediate sell). Without transaction_id, no combination of the other attributes is guaranteed to be unique.

Closure of {transaction_id}

1.	transaction_id → transaction_id	Reflexivity (A1)
2.	transaction_id → $C_T \setminus \{\text{transaction_id}\}$	Given (FD_4)
3.	transaction_id → C_T	Union (U) of 1 and 2

$$\{\text{transaction_id}\}^+ = C_T$$

So {transaction_id} is a superkey.

Minimality

The only proper subset of {transaction_id} is $\emptyset$, and $\emptyset^+ = \emptyset \neq C_T$. So {transaction_id} is a candidate key.

No Other Candidate Keys

{transaction_id} is the only FD left-hand side. Any set X without transaction_id has $X^+ = X \neq C_T$ by the no-fire lemma. Every superkey therefore contains transaction_id.

Candidate key: {transaction_id} (the primary key).

Superkeys ($2^6 = 64$): $\{\text{transaction_id}\} \cup Y$ for every $Y \subseteq C_T \setminus \{\text{transaction_id}\}$.

5 holds

$$C_H = \{\text{user_id}, \text{ticker}, \text{quantity}, \text{average_price}\}$$

FD Set F_H

- FD_5: user_id, ticker $\to$ quantity, average_price. A user has one position in each stock.

Dependencies that do not hold (counterexamples from `sql/seed.sql`):

- user_id $\not\to$ quantity: user 1 holds 2 shares of AAPL and 74 shares of GOOGL.
- ticker $\not\to$ quantity: AAPL is held by user 1 (2 shares), user 5 (12 shares) and user 8 (33 shares).

Closure of {user_id, ticker}

1.	user_id, ticker $\to$ user_id, ticker	Reflexivity (A1)
2.	user_id, ticker $\to$ quantity, average_price	Given (FD_5)
3.	user_id, ticker $\to$ user_id, ticker, quantity, average_price	Union (U) of 1 and 2

$$\{\text{user_id}, \text{ticker}\}^+ = C_H$$

So {user_id, ticker} is a superkey.

Minimality

The proper subsets of {user_id, ticker} are $\emptyset$, {user_id} and {ticker}. None of them contains the left-hand side of FD_5, so by the no-fire lemma:

$$\emptyset^+ = \emptyset \neq C_H$$
$$\{\text{user_id}\}^+ = \{\text{user_id}\} \neq C_H$$
$$\{\text{ticker}\}^+ = \{\text{ticker}\} \neq C_H$$

No proper subset is a superkey, so {user_id, ticker} is a candidate key. The FD user_id $\to$ name, email belongs to `user`, not to F_H, and its right-hand side is not in C_H, so it adds nothing to $\{\text{user_id}\}^+$ here.

No Other Candidate Keys

{user_id, ticker} is the only FD left-hand side. Any set X missing user_id or ticker has $X^+ = X \neq C_H$. Every superkey therefore contains both.

Candidate key: {user_id, ticker} (the composite primary key).
Superkeys ($2^2 = 4$): {user_id, ticker} $\cup\, Y$ for every $Y \subseteq$ {quantity, average_price}.

6 Summary

Relation	FDs	Candidate keys	Superkeys
`user`	FD_1, FD_2	{user_id}, {email}	6
`stock`	FD_3	{ticker}	16
`transaction`	FD_4	{transaction_id}	64
`holds`	FD_5	{user_id, ticker}	4

In every relation, the left-hand side of each FD is a candidate key, so it is a superkey. Every relation is therefore in BCNF.